



4814 - Understanding galaxy formation at cosmic dawn

Cycle: 3, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Xuejian Shen (PI)	Massachusetts Institute of Technology
Prof. Mark Vogelsberger (CoI) (CoPI) (US Admin CoI) (Contact)	Massachusetts Institute of Technology

OBSERVATIONS

ABSTRACT

There are many unresolved questions in our understanding of cosmic dawn ($z\sim 10-30$) and the galaxies illuminating the Universe during this early epoch. This previously unexplored territory has recently been uncovered through early-science observations from JWST, with the discovery of many photometric drop-out galaxy candidates at $z>\sim 10$. These abundant, UV-bright, galaxies have created tensions with our current understanding of galaxy formation within the standard cosmological paradigm, but many uncertainties remain. While this UV abundance tension can potentially be attributed to observational uncertainties, it currently implies that there are missing ingredients in our galaxy formation and cosmological models. With our extensive experience developing theoretical galaxy formation models, we will use state-of-the-art cosmological simulations to expand our understanding of this critical piece of the puzzle. We will investigate the systematic uncertainties inherent in observational estimations, explore the physical causes of discrepancies between theoretical predictions and observations, and suggest methods to empirically test these hypotheses and predictions within existing and novel observations.

OBSERVING DESCRIPTION